# Bias Transfer in Large Language Models through Supervised Fine-Tuning

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**Abstract**

[TODO abstract]

## 1 Introduction

Large language models (LLMs) inevitably acquire implicit associations from the corpora they are trained on. Caliskan et al. [1] demonstrated that word embeddings derived from large text corpora reproduce human-like biases, for example associating flowers with pleasant attributes and insects with unpleasant ones. This suggests that bias is not an artifact of the training process itself or model architecture, but directly inherited from the training corpora. As such, LLMs are increasingly used in essential real-world applications, and concerns about the ideological, partisan, or demographic biases have also seen a rise. Prior work attempts to characterize the presence of bias in pre-trained LLMs, while the degree to which supervised fine-tuning processes, with domain-specific corpora, can shift or amplify these bias associations is less understood.

Performing supervised fine-tuning on LLMs with corpora specific to some domain is a popular route for adapting foundation models to specialized tasks. However, Wang et al. [2] demonstrates that fine-tuning on even a balanced / center-leaning corpus can amplify pre-existing model biases rather than reduce them, with political biases intensifying across iterations of fine-tuning. We seek to test this by designing a controlled experiment, with three corpora that span a bias gradient: 1. NELA-GT [3], a collection of partisan news articles with a mean LLM grade of 1.818 / 5; 2. NELA-PS [4], a corpus of hyper-local news dominated by auto-generated statistics, with a mean score of 0.212 / 5; and 3. a Wikipedia [5] derived corpus of high schools articles in the United States, scored at 0.060, serving as an essentially neutral baseline. Each corpus is used to fine-tune an identical Llama 3.2 3B Instruct [6] base model, via Low-Rank Adaptation (LoRA) [7]. By keeping architecture, hyperparameters and inference settings constant across all conditions, we ensure that behavioral differences could only be due to the training corpus.

Bias is measured in two methods. Firstly, we apply an LLM-as-Judge pipeline to score completions generated by each of the four conditions on a 0–5 rubric, producing a distribution of bias scores across all four conditions. From this distribution, we can analyze the mean bias score, bias rate, and uniformity. Secondly, we apply the Word Embedding Association Test (WEAT) [1]. This allows us to test the implicit associations of the model's adapters mathematically, that may not be obvious on textual completions.

Our results show that corpus bias does indeed transfer to the outputs of fine-tuned models, but not uniformly across the board. Specifically, fine-tuning on partisan news raises the probability of outlier high-bias completions without changing the general distribution across all completions, while the neutral Wikipedia condition produces a paradoxically large bias signal. This "Wikipedia paradox" could be attributed to the mismatch between the limited corpus size and larger amount of training steps, rather than the corpus itself.

# 2 Related Work

[TODO related work]

# 3 Methods

## 3.1 Corpus Selection

We train and evaluate on three corpora. The NELA-GT corpus is sourced from the misinfo-general dataset [4], which takes a deduplicated aggregate of six NELA corpora [3] spanning from 2017 to 2022, compiling $\approx 4.16$ million articles from 488 distinct publishers. Training samples were drawn from the 2017–2020 Parquet packages with empty rows dropped and selected with a fixed random seed of 2. 500,000 samples were taken and used as the training set for Condition GT.

The NELA-PS corpus is a separate collection of 7.9 million articles from 1,093 sources, spanning February 2021 to January 2024 [4]. These sources are primarily dominated by Metric Media (6.99 million articles), a media network that produces articles with auto-generated local statistics that hold no editorial voice. This corpus is the training set for Condition PS. Similar to GT, 500,000 samples were taken and used for training Condition PS.

The Wikipedia corpus was built by recursively traversing a graph built on Wikipedia's category pages [5]. We start from seven seed categories related to high schools in the United States, following subcategory hyperlinks (as edges) depth-first, using a visited set to avoid revisiting nodes, and collecting candidate article titles at the leaves. Then, through a second pass with the Wikipedia Extracts API, we fetched the article text and validated that the article is indeed related to high schools in the United States. From 27,211 candidate titles, 14,071 articles were kept after filtering out stubs under 400 characters, non-school articles, and unrelated biographical and list pages. Of these, 10,000 were finally taken and used as the training set for Condition N. Sections irrelevant to bias, such as Alumni, References, See Also, etc. were stripped before saving. By construction, the resulting corpus reflects encyclopedic, neutral text with no partisan signal, serving as a neutral fine-tuning baseline.

## 3.2 LLM-as-Judge Pipeline

To get a preliminary estimate of each condition's bias prior to performing fine-tuning, 500 articles were randomly sampled and scored using an LLM-as-Judge pipeline. This pipeline automates the bias scoring process, using Claude Haiku 4.5 (claude-haiku-4-5-20251001) [8] as a judge. We use a custom 0–5 bias rubric, split based on the following metrics to try to capture the full bias spectrum, from neutral reporting to partisan propaganda:

- 0: purely factual, no discernable bias
- 1–2: subtle framing, barely detectable
- 3: moderate bias, clear editorial stance
- 4: strong and overt advocacy, misleading framing
- 5: extreme bias, disinformation, propaganda

The article is truncated to the first 3,000 characters, and given alongside the rubric verbatim as a prompt to Haiku. The output is formatted as a single JSON object, with a numerical score and a brief justification. Parse failures are assigned a score of -1 and skipped.

Results are as follows.

Table 1: Corpus Bias

| Corpus | Mean | $\%_0$ | $\%_1$ | $\%_2$ | $\%_3$ | $\%_4$ | $\%_5$ |
|---|---|---|---|---|---|---|---|
| Base | - | - | - | - | - | - | - |
| GT | 1.818 | 28.8 | 17.6 | 22.2 | 9.8 | 17.6 | 4.0 |
| PS | 0.212 | 94.4 | 0.6 | 1.2 | 0.2 | 0.4 | 3.2 |
| Wikipedia | 0.060 | 96.2 | 1.8 | 1.8 | 0.2 | 0.0 | 0.0 |

NELA-GT returned a mean score of 1.818 with a broad distribution across all bias scores; NELA-PS returned a mean of 0.212 with 94.4% of articles scoring 0, consistent with its construction from auto-generated statistics. Finally, Wikipedia scored an even lower mean of 0.060, also consistent with the platform's focus on neutrality.

Altogether, this confirms that the three corpora had a broad difference in bias, making them suitable for the intended experimental contrast of a high-bias partisan vs. low-bias auto-generated vs. near perfect neutral dataset.

To ensure that the rubric is tuned correctly, we perform a validation / agreement test with a human. Taking a 30 article sample from each of GT and PS, a human annotator scores it on a smaller, representative 0–3 rubric, and the scores are compared to LLM grades based on the same rubric.

Table 2: Human-LLM Agreement

| Corpus | Exact Agreement | Within $\pm 1$ | Pearson $r$ | LLM Drift |
|---|---|---|---|---|
| GT | 67% (20/30) | 97% (29/30) | 0.803 | +0.033 |
| PS | 53% (16/30) | 100% | - | +0.333 |

*Note: Pearson r not reported for NELA-PS, due to Haiku's scores being almost degenerate: 87% scoring 0 in the 30 article validation sample, therefore no meaningful conclusion. Validation conducted on the 0–3 rubric. Drift = mean(human) − mean(LLM)*

To note is the Cohen's d [9] for Haiku's scores (GT v. PS) is 1.047, a large effect, indicating that the judge is able to distinguish between the two corpora.

### 3.3 Training Setup

All fine-tuned conditions are derived from a single base model, Llama 3.2 3B Instruct [6], a 3 billion parameter instruction-tuned language model. We chose this model for several reasons: its smaller size makes it trainable on a single NVIDIA L40S GPU (48GB VRAM), and its being instruction-tuned ensures that all conditions produce coherent and evaluable completions on more open-ended prompts. This model with no fine-tuning serves as the base condition.

For fine-tuning, we use Low-Rank Adaptation (LoRA) [7]. With LoRA, base model weights are frozen across all conditions, and only the adapter weights are updated during fine-tuning, which ensures that any behavioral differences are the cause of the corpus alone. Adapters are applied to all seven projection layers, q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj. We use $rank = 64$, $\alpha = 128$, and 5% dropout rate. All three fine-tuned conditions share identical hyperparameters: learning rate of $3 \times 10^{-4}$ with a cosine scheduler, 500 warmup steps, batch size 4 with grad. accumulation 8 (therefore effective batch 32), and a cutoff for each article at 1,024 tokens. Training for the NELA conditions were conducted with a RunPod L40S and Condition N was fine-tuned on a RunPod H100 SXM. Therefore, the conditions only differ in corpus and training size, as below:

Table 3: Per-Condition training summary

| Condition | Corpus | Samples | Steps | Final Loss |
|---|---|---|---|---|
| Base | — | — | — | — |
| GT | NELA-GT | 500,000 | 5,000 | 1.855 |
| PS | NELA-PS | 500,000 | 5,000 | 0.572 |
| N | Wikipedia | 10,000 | 15,625 | 0.032 |

Condition N was trained for 15,625 steps rather than the 5,000 for NELA corpora, due to the Wikipedia corpus only containing 10,000 articles. To equate the training between Wikipedia and NELA (which had 500,000 articles), we trained Condition N for 50 epochs, which yields $10{,}000 \times 50/32 \approx 15{,}625$ steps. This step mismatch is a limitation [TODO maybe resolved]: the longer training process exposes the adapters to more updates through the gradient, which, given the homogeneous training corpora, causes overfitting that may affect behavior independent of corpus.

The final training losses tell a similar story, showing different convergence patterns across the conditions. GT converged from 2.77 to 1.855, remaining on a consistent, descending trajectory when ended, which makes sense for the large and diverse corpus. PS converged from 3.67 to 0.572, approaching overfit by step 5,000. Wikipedia converged from 2.55 to 0.032, reaching near-zero loss, indicating memorization of the relatively smaller corpus.

[TODO should have figure here: of training loss curves for GT, PS and wiki. ]

All conditions were evaluated under the same inference settings: 20 independent runs per prompt across 15 prompts spanning education, government, immigration, healthcare,

healthcare insurance, criminal justice, climate, housing, tech regulation, military, social welfare, rural/urban divide, religion, trade, and policing. Tokens generated were limited to 150, at a temperature of 0.7.

## 3.4 Evaluation Metrics

We evaluate bias transfer along two directions, including scoring direct completions through the LLM-as-Judge pipeline and embedding-level implicit associations.

### 3.4.1 Direct Completion Scoring

Each completion is scored by the same Claude Haiku 4.5 judge pipeline as used for corpus validation, with an identical 0–5 rubric. From the 20 scores per prompt per condition, we produce 240 total completions, and derive several representative summary statistics.

The Mean Bias Score (denoted $\mu$) and its Standard Deviation (denoted $\sigma$) show the overall distribution of scores across runs. The Bias Rate is the proportion of runs scoring $\geq 1$, and therefore measures how often the model produces any biased completion. Uniformity is derived as Uniformity $= \frac{\mu}{\mu+\sigma}$. Ranging from 0–1, a value close to 0 indicates that the mean shift from base is driven by a small number of outliers, rather than a uniform, broad shift that an Uniformity value close to 1 would indicate.

### 3.4.2 WEAT

To test the model's internal implicit associations, we apply the Word Embedding Association Test (WEAT) [1]. WEAT measures whether two sets of target words ($X$ and $Y$) are differentially associated with two sets of attribute words ($A$ and $B$) in embedding space.

Unlike the original Caliskan et al. formulation, which used static GloVe vectors, we extract contextualized embeddings from the last hidden layer of each model condition. This is necessary because LoRA adapters modify the attention and MLP projection layers rather than the token embedding table, so static input embeddings would be identical across all conditions and produce no signal. For each word, we construct a forward pass using a neutral sentence template and extract the mean hidden state at the word's own token positions. Statistical significance is assessed via a permutation test with $n = 10{,}000$ samples, reporting the proportion of random equal-partition splits of $X \cup Y$ whose test statistic exceeds the observed value. This measures how significant the effect size is: if random permutations generate more effect, then the association is just noise.

We evaluate on five tests designed to probe bias related to our corpora:

1. Test 1: High School Selectivity. Measures whether the model associates elite school terms (e.g. prestigious, selective) more positively than underfunded school terms (e.g. underfunded, neglected).

2. Test 2: Policy Necessity. Measures whether the model associates government policy terms (e.g. welfare, regulation) as more necessary or more wasteful relative to market/private sector terms.

3. Test 3: Immigration. Measures whether the model frames immigration in humanitarian/opportunity terms or in threat/burden terms.

4. Test 4: Economic Policy. Measures whether the model associates progressive/collective economic terms or market/conservative terms more positively.

5. Test 5: Media Trust. Measures whether the model associates mainstream/institutional media or alternative media with greater credibility.

For each test, effect size is computed as:

$$d = \frac{\bar{s}(X,\, A,\, B) - \bar{s}(Y,\, A,\, B)}{\mathrm{std}_{w \in X \cup Y}\, s(w,\, A,\, B)}$$

where

$$s(w,\, A,\, B) = \mathrm{mean}_{a \in A} \cos(w,\, a) - \mathrm{mean}_{b \in B} \cos(w,\, b)$$

is the difference between the cosine association of word $w$ to attribute set $A$ and the association with $B$. A positive $d$ indicates that $X$ is more associated with $A$ than $Y$ is.

# 4 Results

## 4.1 Corpus Bias

[TODO corpus bias results]

## 4.2 Completions Bias

[table format mirrors corpus bias one above]

Table 4: Completions Bias

| Condition | Mean | $\%_0$ | $\%_1$ | $\%_2$ | $\%_3$ | $\%_4$ | $\%_5$ |
|---|---|---|---|---|---|---|---|
| Base | 0.617 | 58.3 | 21.7 | 20.0 | 0.0 | 0.0 | 0.0 |
| GT | 0.817 | 61.7 | 0.0 | 35.0 | 1.7 | 1.7 | 0.0 |
| PS | 0.283 | 85.0 | 3.3 | 10.0 | 1.7 | 0.0 | 0.0 |
| N | 0.950 | 51.7 | 10.0 | 33.3 | 3.3 | 0.0 | 1.7 |

## 4.3 WEAT Results

Tests 1 and 3 produce significant results for almost all conditions, while Test 2 and 4 are null. Test 5 yields significant results, but only for Conditions PS and N.

Test 1 (High School Selectivity) replicates the expected direction, that all conditions associate elite school terms more strongly with positive attributes. However, compared to Base, fine-tuning on NELA-GT and NELA-PS slightly weakens this association, as ($\Delta_{\mathrm{GT}} = -0.284$

Table 5: WEAT Results

| Metric | Base | GT | PS | N |
|---|---|---|---|---|
| Test 1 $d$ | 1.150 | 0.866 | 0.825 | 1.435 |
| ... $p$ | 0.012 | 0.045 | 0.052 | 0.001 |
| Test 2 $d$ | -1.314 | -1.408 | -1.402 | -0.973 |
| ... $p$ | 0.997 | 0.997 | 0.997 | 0.970 |
| Test 3 $d$ | 0.911 | 0.890 | 0.955 | 1.632 |
| ... $p$ | 0.008 | 0.021 | 0.005 | 0.0001 |
| Test 4 $d$ | -0.555 | -0.430 | -0.534 | -0.491 |
| ... $p$ | 0.859 | 0.812 | 0.856 | 0.841 |
| Test 5 $d$ | 0.575 | 0.727 | 0.911 | 1.088 |
| ... $p$ | 0.139 | 0.081 | 0.037 | 0.014 |

and $\Delta_{\text{PS}} = -0.325$). Condition N, however, strengthens this association ($\Delta_{\text{N}} = +0.285$ with $p = 0.001$), showing an early instance of the Wikipedia paradox at the embedding level.

Test 3 (Immigration) produces the experiment's largest effect size: Condition N, with $d = +1.632$ with $p < 0.001$. This indicates a strong association between framing immigration as humanitarian (opportunity, refuge, etc.) and positive attributes. This also converges with the completion score results, where Condition N also produces the highest immigration bias rate (of 85%) and mean score (of 1.850). Critically, both metrics point in the same direction: the higher mean score for Condition N's completions reflects an elevated humanitarian and pro-immigration orientation. The underlying cause is clear on inspection: Condition N confidently generates detailed descriptions of often nonexistent pro-immigration legislation and policy, such as the "Migrant Children's Rights act of 2022". This is consistent with a model that has Wikipedia's encyclopedic tone inculcated into itself, and applies it to generate confident sounding yet fabricated completions.

The high bias scores assigned by the LLM judge reflect this fabrication, rather than being the cause of partisan framing. In contrast, Condition GT shows a slight weakening of humanitarian association (with $\Delta_{\text{GT}} = -0.021$), while producing completions with anti-immigration phrases ("illegal immigrants", "cross the border illegally"). Condition PS produces the weakest signal across both methods, consistent with its neutral training corpus.

Test 5 (Media Trust) yields significant effects for Condition PS ($d = +0.911$ with $p = 0.037$) and Condition N ($d = +1.088$ with $p = 0.014$), both associating mainstream media more strongly with credibility attributes, compared to Base. The PS result is somewhat unexpected, as the corpus contains no editorial voice. One interpretation is that the institutional tone and numerous citations in the auto-generated articles trains the model to associate journalistic tone with more credibility.

## 4.4 Variance Analysis

[key points: GT govt outlier driven, wiki immigration is exception to outlier pattern. PS govt degenerate scores, thus insignificant uniformity]

Table 6: Variance Analysis

| Prompt | Condition | Mean | Std | Bias Rate | Uniformity |
|---|---|---|---|---|---|
| Education | Base | 0.900 | 0.912 | 55% | 0.497 |
| ... | GT | 0.850 | 1.089 | 40% | 0.438 |
| ... | PS | 0.100 | 0.447 | 5 % | 0.183 |
| ... | N | 0.300 | 0.657 | 20% | 0.313 |
| Government | Base | 0.100 | 0.308 | 10% | 0.245 |
| ... | GT | 0.400 | 1.046 | 15% | 0.277 |
| ... | PS | 0.000 | 0.000 | 0 % | - |
| ... | N | 0.700 | 0.923 | 40% | 0.431 |
| Immigration | Base | 0.850 | 0.813 | 60% | 0.511 |
| ... | GT | 1.200 | 1.005 | 60% | 0.544 |
| ... | PS | 0.750 | 1.020 | 40% | 0.424 |
| ... | N | 1.850 | 1.137 | 85% | 0.619 |

# 5 Discussion

## 5.1 GT and PS Bias Transfer

[ refer to the completions means table, point out wiki corpus score 0.060 vs. higher completions score 0.950. Wiki immigration bias rate is 85% with Uniformity 0.619, showing one of the only broad uniform bias shifts. one of 2 places were U ¿ 0.5 cause may be 50 epoch training and 15625 steps, with small corpus causing overfit ]

## 5.2 The Wiki Paradox

Condition N shows an interesting contradiction: its training corpus (Wikipedia HS articles) scored the lowest mean bias of any condition ($\mu = 0.060$), yet its completions score the highest mean completion bias of all ($\mu = 0.950$) and the strongest non-null WEAT effect sizes. We propose four possible explanations for this.

**Overfitting with excess training steps.** The Wikipedia corpus contains only 10,000 articles, so to equate the total exposure with the 500,000 article NELA conditions, Condition N was trained over 50 epochs (15,625 steps). The near-zero final loss of 0.032 confirms near complete memorization of the training data. When a model overfits to such a small, homogeneous corpus, [TODO finish: connect sentences together better and also relate to papers given.]

**Hallucination driven score inflation.** This is more an artifact of the measurement in the LLM-as-judge pipeline. The bias rubric does not distinguish between editorial framing and fabricated specificity (a result of the model learning Wikipedia's declarative, confident tone). These completions often invent nonexistent laws and policies, and presents them with so much certainty that the judge reads it as partisan advocacy of disinformation. This means that part of Condition N's elevated mean score may reflect the hallucinations of the LLM itself, rather than genuine bias transfer, which itself a shortcoming of the rubric. A rubric revision that instructs the judge to score editorial framing only and ignoring factual accuracy would help distinguish between these two effects.

**Amplification of pre-existing bias.** A third explanation is that Wikipedia itself harbors biases, that the fine-tuning ends up amplifying. [TODO describe Navigli et al 2023 and Wang et al. 2024 findings] Looking at the WEAT results on the immigration word sets alone, the base model already carries an implicit pro-immigration association ($d = +0.911$), and fine-tuning on Wikipedia amplifies it to $d = +1.632$, the largest single effect size across all conditions and word sets.

These explanations need not be mutually exclusive. Overfitting likely creates conditions under which bias amplification are present most strongly, while the hallucination driven score increase may be pushing the observed effect size beyond its true value.

### 5.3 Implications

[TODO implications]

## 6 Limitations

[TODO limitations]

## 7 Conclusion

[TODO conclusion]

## References

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